

UNIVERSITY OF CALGARY
FACULTY OF SCIENCE
DEPARTMENT OF MATHEMATICS & STATISTICS**FINAL EXAMINATION**
LINEAR METHODS I – MATH 211
ALL SECTIONS – FALL 2014

Date: December 11, 2014

Time: 3 hours

STUDENT I.D. NUMBER	LAST NAME	FIRST NAME

EXAMINATION RULES

1. This is a closed book examination.
2. No aids are allowed for this examination.
3. Answer Part A on the scantron sheet and Part B on the examination paper.
4. Scantron sheets must be filled in during the exam time period. No additional time will be granted to fill in the scantron form.
5. The use of personal electronic or communication devices is prohibited.
6. A University of Calgary Student ID card is required to write the Final Examination and could be requested for midterm examinations. If adequate ID isn't presented, the student must complete an Identification Form.
7. Students late in arriving will not be permitted after one-half hour of the examination time has passed.
8. No student will be permitted to leave the examination room during the first 30 minutes, nor during the last 15 minutes of the examination. Students must stop writing and hand in their exam immediately when time expires.
9. All inquiries and requests must be addressed to the exam supervisor.
10. Students are strictly cautioned against:
 - (a) communicating to other students;
 - (b) leaving answer papers exposed to view;
 - (c) attempting to read other students' examination papers.
11. During the final examination, if a student becomes ill or receives word of domestic affliction, the student must report to the Invigilator, hand in the unfinished paper and request that it be cancelled. If ill, the student must report immediately to a physician/counselor for a medical note to support a deferred examination application.
12. Once the examination has been handed in for marking, a student cannot request that the examination be cancelled. Retroactive withdrawals from the course will be denied.
13. Failure to comply with these regulations will result in rejection of the examination paper.

Question	Total Marks	Actual Marks
B1	5	
B2	6	
B3	7	
Total	18	

Part A: Mark your answers on the scantron sheet provided.

Each question is worth 2 points.

1. Consider the following augmented matrix in which * denotes an arbitrary number and ■ denotes a **nonzero** number.

$$\left[\begin{array}{cccc|c} \blacksquare & * & * & * & \blacksquare \\ 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

What can you say about the corresponding system of linear equations?

- (a) It has no solution.
 - (b) It has a unique solution.
 - (c) The general solution has exactly one parameter.
 - ☒ (d) The general solution has exactly two parameters.
 - (e) There is insufficient information to determine the answer.
2. For which values of a and b does the system

$$\begin{aligned} ax + 2y &= 5 \\ x + by &= -1 \end{aligned}$$

have infinitely many solutions?

- (a) $a = -5$ and $b \neq -2/5$.
- (b) $a \neq -5$ and $b \neq -2/5$.
- ☒ (c) $a = -5$ and $b = -2/5$.
- (d) $a \neq -5$ and $b = -2/5$.
- (e) None of the above.

$$\begin{aligned} 5 + a &= 0 \\ a &= -5 \end{aligned}$$

$$\begin{aligned} 2 - 2ab &= 0 \\ b &= 2/a = -2/5 \end{aligned}$$

$$\begin{aligned} &\begin{bmatrix} a & 2 & 5 \\ 1 & b & -1 \end{bmatrix} \\ &\downarrow \\ &\begin{bmatrix} 1 & b & -1 \\ a & 2 & 5 \end{bmatrix} \\ &\downarrow \\ &\begin{bmatrix} 1 & b & -1 \\ 0 & 2-ab & 5+a \end{bmatrix} \end{aligned}$$

3. Let A be the augmented matrix and C the coefficient matrix for a system of m linear equations in n variables. If $\text{rank}(A) = \text{rank}(C) = n$, which of the following statements must be true?

- (a) The system has no solution.
- ☒ (b) The system has a unique solution.
- (c) The general solution must have at exactly one parameter.
- (d) The general solution has at least two parameters.
- ☒ (e) There is insufficient information to determine the answer.

$$\begin{array}{c} n \\ \begin{bmatrix} C & | & \end{bmatrix} \\ m \\ \underbrace{\hspace{10em}}_A \end{array}$$

4. Let

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} \text{ and } C = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

If $AYB = C$, then Y is equal to

(a) $\begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix}.$

(b) $\begin{bmatrix} -2 & 2 \\ 1 & -1 \end{bmatrix}.$

(c) $\begin{bmatrix} 3 & -2 \\ -3 & 2 \end{bmatrix}.$

(d) $\begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix}.$

(e) none of the above.

$$\overline{A}^{-1} A Y B \overline{B}^{-1} = \overline{A}^{-1} C \overline{B}^{-1}$$

$$Y = \overline{A}^{-1} C \overline{B}^{-1}$$

$$= \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -2 \\ -3 & 2 \end{bmatrix}$$

5. Suppose $A = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$ and $AB = \begin{bmatrix} 1 & 1 & 0 & 3 \\ -1 & 4 & 2 & 5 \end{bmatrix}$. The $(1,4)$ -entry of B is

(a) -4.

(b) -2.

(c) 0.

(d) 3.

(e) 7.

$$B = \overline{A}^{-1} (AB)$$

$$= \begin{bmatrix} 4 & -1 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 3 \\ -1 & 4 & 2 & 5 \end{bmatrix}$$

$$(1,4) \text{ entry } 12 - 5 = +7$$

6. Let A , B , and C be 2×2 matrices with $\det(A) = 2$, $\det(B) = 4$, and $\det(C) = -2$. Then $\det(2(A^T)^3(C^{-1} + \text{adj}C)B^{-1})$ is equal to

(a) 2.

(b) -2.

(c) 4.

(d) -4.

(e) none of the above.

$$= 2^2 \det(A)^3 (\det(C^{-1} + \det(C) C^{-1}) \frac{1}{\det B})$$

$$= \frac{32}{4} (1 + \det(C))^2 \det(C)^{-1}$$

$$= 8 \frac{(-1)^2}{-2} = -4$$

7. Find the matrix C so that $\text{adj}(C) = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$. $= [\text{cof}(C)]^T$
- (a) $C = \begin{bmatrix} 4 & -2 \\ -1 & 1 \end{bmatrix}$. $\text{adj}(C) = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}^T = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix} \times$
- (b) $C = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$. $\text{adj}(C) = \frac{1}{2} \begin{bmatrix} 4 & -2 \\ -1 & 1 \end{bmatrix}^T = \frac{1}{2} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix} \times$
- (c) $C = 2 \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$. $\text{adj}(C) = 2 \begin{bmatrix} 4 & -2 \\ -1 & 1 \end{bmatrix}^T = 2 \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix} \times$
- (d) $C = \frac{1}{2} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix}$. $\text{adj}(C) = \frac{1}{2} \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}^T = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \times$
- (e) None of the above.

8. Find the determinant of $\begin{bmatrix} i & i & 0 \\ 1 & i & 0 \\ 0 & \pi & -1 \end{bmatrix}$.

(a) $1 + i$.

(b) $1 - i$.

(c) $-1 + i$.

(d) $-1 - i$.

(e) None of the above.

along column 3 =
 $0 + 0 + (-1)[i^3 - i]$
 $= -(-1 - i)$
 $= +1 + i$

9. Parametric equations of a line are given as

$$x = 2t - 1$$

$$y = 2 - 4t$$

$$z = 2 - 2t$$

any multiply $\vec{d} = \begin{bmatrix} 2 \\ -4 \\ -2 \end{bmatrix}$

A direction vector for the line is

(a) $\begin{bmatrix} -1 \\ -4 \\ -2 \end{bmatrix}$

(b) $\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$

(c) $\begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$

(d) $\begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$

(e) $\begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}$

10. Let $\vec{u} = [1, 3, 5]^T$ and $\vec{v} = [2, 1, -1]^T$. Which of the following statements is true?

(a) the angle between $\vec{u}$ and $\vec{v}$ is acute.

(b) the angle between $\vec{u}$ and $\vec{v}$ is obtuse.

(c) $\vec{u}$ and $\vec{v}$ are perpendicular.

(d) $\vec{u}$ and $\vec{v}$ are parallel.

(e) None of the above.

$$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|} = \frac{2+3-5}{\|\vec{u}\| \|\vec{v}\|} = \frac{0}{\|\vec{u}\| \|\vec{v}\|} = 0$$

11. The shortest distance from the plane with equation $x - y + 2z = 5$ to the plane with equation $2x - 2y + 4z = 4$ is:

(a) 0.

(b) 1.

(c) $\sqrt{6}/2$.

(d) $\sqrt{6}$.

(e) None of the above.

Planes are parallel

$P_0 = (2, 0, 0)$ on first plane

$P_1 = (5, 0, 1)$ on second plane.

Shortest distance

$$\|\vec{P_0 P_1}\| = \left| \frac{\vec{P_0 P_1} \cdot \vec{n}}{\|\vec{n}\|} \right| = \frac{3}{\sqrt{6}} = \frac{3\sqrt{6}}{6} = \frac{\sqrt{6}}{2}$$

$\vec{n} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$ $\vec{P_0 P_1} = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$

12. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation such that

$$T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \text{and} \quad T \begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Find $T^{-1} \left(\begin{bmatrix} 1 \\ 2 \end{bmatrix} \right)$.

(a) $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$

(b) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(d) $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

(e) $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}, \quad \text{so} \quad T^{-1} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$T^{-1} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

13. Let $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be linear transformations such that

$$S \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}, \quad S \begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \quad T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad T \begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Find $(S \circ T) \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)$. $= S(T[1]) = S[-1]$

(a) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

(c) $\begin{bmatrix} 0 \\ -2 \end{bmatrix}$

(d) $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$

(e) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} -1 \\ -1 \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 0 \\ -2 \end{bmatrix} \Rightarrow \begin{matrix} a=1 \\ b=1 \end{matrix} \quad S \begin{bmatrix} -1 \\ -1 \end{bmatrix} = S \begin{bmatrix} 1 \\ 1 \end{bmatrix} + S \begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

14. The expression

$$\frac{5+5i}{1+3i} + \frac{4+7i}{2-3i}$$

simplifies to

(a) $-1+i$.

(b) $-1-i$.

(c) $1+i$.

(d) $1-i$.

(e) none of the above.

$$= \frac{(5+5i)(2-3i) + (4+7i)(1-3i)}{(1+3i)(2-3i)}$$

$$= \frac{10-15i+10i-15i^2+4+12i+7i+21i^2}{2-3i+6i-9i^2}$$

$$= \frac{8+14i}{11+3i} \times \frac{11-3i}{11-3i} = \frac{88-24i+154i-42i^2}{121+9} = \frac{130+130i}{130} = 1+i$$

15. The matrix of the linear transformation that results from reflection in the line $y = -x$, followed by rotation through an angle of $\pi/6$, followed by reflection in the y axis is

(a) $\begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}$.

(b) $\begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix}$.

(c) $\begin{bmatrix} 1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & 1/2 \end{bmatrix}$.

(d) $\begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}$.

(e) none of the above.

matrix of $y = -x$ $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

rot $\pi/6$

$$\begin{bmatrix} \cos \pi/6 & -\sin \pi/6 \\ \sin \pi/6 & \cos \pi/6 \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix}$$

y -axis

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

so result is

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} -\sqrt{3}/2 & 1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}$$

16. Suppose $p(x) = x^2 - ix + c$ is a polynomial with $-i$ as a root. What is the value of c ?

(a) $c = 0$.

(b) $c = 2$.

(c) $c = -2$.

(d) $c = 2i$.

(e) $c = -2i$.

$$\begin{aligned} (-i)^2 - i(-i) + c &= 0 \\ -1 - 1 + c &= 0 \\ c &= +2 \end{aligned}$$

17. If $z = -1 + \sqrt{3}i$, then the argument of z^{90} is equal to

(a) 0.

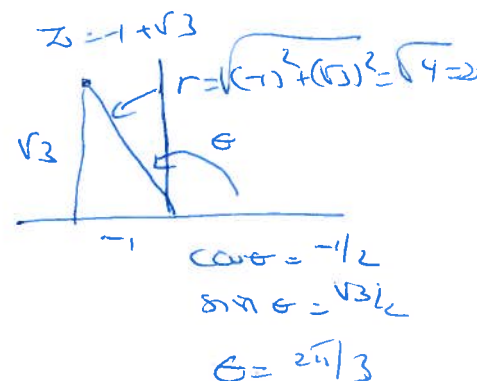
(b) $\pi/3$.

(c) $2\pi/3$.

(d) $4\pi/3$.

(e) none of the above.

$$\begin{aligned} z &= 2e^{i\frac{2\pi}{3}} \\ z^{90} &= 2^{90} e^{i\frac{2\pi}{3} \times 90} \\ &= 2^{90} e^{i60\pi} \\ &= 2^{90} e^{i0} \\ &= 2^{90} \end{aligned}$$



18. The matrix A can be written in the form

$$A = \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} -2/3 & 1/3 \\ 1/3 & 1/3 \end{bmatrix} = P D P^{-1}$$

Which of the following statements **must** be false?

✓ (a) The matrix A is diagonalizable.

✓ (b) $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ is an eigenvector for A .

(c) For any integer $n \geq 2$, we can write $A^n = 3^{n-1} \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$.

✗ (d) The characteristic polynomial of A is $c_A(x) = x(x+3)$.

✓ (e) A has distinct eigenvalues.

$$(0 \text{ and } 3) \rightarrow c_A(x) = x(x-3)$$

19. Let $A = \begin{bmatrix} 1 & \square & 3 \\ \square & \square & \square \\ 1 & \square & -1 \end{bmatrix}$, where $\square$ are hidden entries of A known only to Professor Proton. If

$X = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$ is an eigenvector of A , then which of the following statements **must** be true?

- (a) -2 is an eigenvalue of A .
- (b) -1 is an eigenvalue of A .
- (c) 0 is an eigenvalue of A .
- (d) 1 is an eigenvalue of A .
- (e) 2 is an eigenvalue of A .

$$AX = \lambda X$$

$$\begin{bmatrix} 6 \\ \\ \end{bmatrix} = \lambda \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

$$\therefore \lambda = 2$$

20. The eigenvalues of $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ -2 & -2 & -1 \end{bmatrix}$ are

- (a) $1, -1$, and -3 .
- (b) $1, 1 - 2i$ and $1 + 2i$.
- (c) 3 and -1 , with -1 having multiplicity two. ~~Correct answer.~~
- (d) 1 and -1 , with 1 having multiplicity two.
- (e) none of the above.

$$C_A(\lambda) = \det \begin{bmatrix} \lambda - 1 & -2 & 0 \\ -2 & \lambda - 1 & 0 \\ 2 & 2 & \lambda + 1 \end{bmatrix}$$

$$= (\lambda + 1)((\lambda - 1)^2 - 4)$$

$$= (\lambda + 1)(\lambda^2 - 2\lambda - 3)$$

$$= (\lambda + 1)(\lambda + 1)(\lambda - 3)$$

21. Consider an $n \times n$ matrix A . Which of the following statements is ALWAYS true?

- (a) A is always diagonalizable.
- (b) If for some λ the general solution of $(\lambda I - A)X = 0$ has infinitely many solutions, then A is diagonalizable.
- (c) If the characteristic polynomial of A has a single root of multiplicity n , then A is diagonalizable.
- (d) If the characteristic polynomial of A has n distinct roots, then A is diagonalizable.
- (e) None of the above.

Part B: Provide your full solution on these pages.

Question 1 (5 points) Let $A = \begin{bmatrix} 1 & 0 & -3 & -2 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

Provide your answers in the box below.

- a) What is the rank of A ?
- b) How many parameters, if any, are there in the general solution to $AX = 0$?
- c) If possible, find a nonzero vector B such that $AX = B$ has no solutions. Otherwise, state that "No such vector exists."
- d) If possible, find a nonzero vector B such that $AX = B$ has a unique solution. Otherwise, state that "No such vector exists."
- e) If possible, find a nonzero vector B such that $AX = B$ has infinitely many solutions. Otherwise, state that "No such vector exists."

(a): 3 (b): 2 (c): $\begin{bmatrix} * \\ * \\ * \\ \text{any} \end{bmatrix}$ (d): NSVE (e): $\begin{bmatrix} * \\ * \\ * \\ 0 \end{bmatrix}$

ANY
NON
ZERO

MUST
BE
0

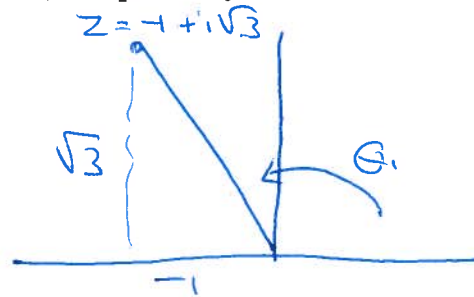
Question 2 (6 points) Write a clear and complete solution, and provide your final answer in the box below:

[2] a) Write $-1 + i\sqrt{3}$ in polar form.

$$r = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{4} = 2$$

$$\cos \theta = -1/2 \quad \sin \theta = \sqrt{3}/2$$

$$\therefore \theta = 2\pi/3$$



$$-1 + i\sqrt{3} = 2e^{i2\pi/3}$$

[4] b) Find all complex numbers z such that $z^4 = 8(-1 + i\sqrt{3})$, and express each in polar form.

$$8(-1 + i\sqrt{3}) = 8 \cdot 2e^{i2\pi/3} = 16e^{i2\pi/3}$$

$$z = re^{i\theta} \rightarrow z^4 = r^4 e^{i4\theta}$$

$$\text{We get } r^4 e^{i4\theta} = 16e^{i2\pi/3}$$

$$\therefore r^4 = 16 \Rightarrow r = 2$$

$$4\theta = \frac{2\pi}{3} + 2k\pi$$

$$\boxed{\theta = \frac{\pi}{6} + \frac{k\pi}{2} \quad k=0,1,2,3}$$

$$k=0 \quad \theta = \pi/6 \quad z_0 = 2e^{i\pi/6}$$

$$k=2 \quad \theta = 7\pi/6 \quad z_2 = 2e^{i7\pi/6}$$

$$k=1 \quad \theta = 2\pi/3 \quad z_1 = 2e^{i2\pi/3}$$

$$k=3 \quad \theta = 5\pi/3 \quad z_3 = 2e^{i5\pi/3}$$

(a):

$$2e^{i2\pi/3}$$

(b):

$$2e^{i\pi/6}, 2e^{i2\pi/3}, 2e^{i7\pi/6}, 2e^{i5\pi/3}$$

Question 3 (7 points) Consider the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -2 \\ 0 & 2 & -1 \end{bmatrix}$.

Write a clear and complete solution, and provide your final answers in the box below:

[2] a) Find the characteristic polynomial of A .

$$\begin{aligned}
 C_A(x) &= \det(xI - A) = \begin{vmatrix} x-1 & 0 & 0 \\ 0 & x-3 & 2 \\ 0 & -2 & x+1 \end{vmatrix} \\
 &= (x-1)((x-3)(x+1) + 4) \\
 &= (x-1)(x^2 + x - 3x - 3 + 4) \\
 &= (x-1)(x^2 - 2x + 1) = (x-1)^3
 \end{aligned}$$

[2] b) Find the eigenvalues of A and their multiplicities.

$\lambda = 1$ only eigenvalue.
multiplicity 3

[3] c) With justification, is the matrix A diagonalizable? (Give the simplest reason possible).

Solve $(\lambda I - A)x = 0$ with $\lambda = 1$.

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & -2 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \text{general solution has 2 parameters}$$

(a): $(x-1)^3$

(b): $\lambda = 1$
multip. 3

(c): NO!
multiplicity is 3 $\neq$
basis eigenvectors 2